UNIVERSITAT JAUME I

HAMLET:

Hand Anthropometric Measurement and Landmark Extraction Tool

Complete User Guide

Purpose of the guide. This document explains in full how to install, launch and use the application in its current version. It is intended to help users who are not familiar with the project understand the complete workflow, the input files, the generated results and the manual review points required to obtain reliable measurements.

Document Information

Field	Value
Application	Hand Anthropometric Measurement and Landmark Extraction Tool
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1. Scope and Overview

Hand Anthropometric Measurement and Landmark Extraction Tool is a desktop application designed to obtain anthropometric hand measurements from a 3D model in OBJ format and its associated texture. The tool combines automatic anatomical marker detection, 2D-to-3D mapping, manual validation, rigging assisted by OpenPose, automatic labeling, interactive correction and final distance calculation.

This guide covers the use of the executable version of the application: installation, file selection, marker detection and review, rigging, labeling, measurement selection, final calculation and reopening previously processed projects. It does not assume prior knowledge of older versions or the hand scanning protocol. A demonstration video illustrating the complete application workflow is provided with the software release. The software release, together with this guide and the accompanying demonstration video, is archived in Zenodo as a citable, versioned resource with a persistent DOI.

2. Requirements and Installation

System Requirements

Requirement	Description
Operating system	64-bit Windows 10 or Windows 11.
CPU	64-bit processor. Performance depends on the size of the mesh and the texture.
RAM	Recommended minimum: 8 GB. Recommended for smooth use: 16 GB or more.
Storage	Application size: 2.5 GB. Additional space required to save results.
Permissions	Use a folder with read/write permissions. Avoid locations blocked by corporate policies.

Installing the Executable Version

1. Download the ZIP file published in the release.
2. Extract the ZIP file to a local folder, such as Documents or a user work folder.
3. Open the complete extracted folder.
4. Run ***HAMLET.exe***.

To transfer the application to another computer, do not copy only the .exe. The portable application needs the folders that accompany the executable, such as assets, Blender, OpenPose, PySide6 and vtkmodules. To move the application to another computer, copy or distribute the complete folder or the complete ZIP file.

3. Input, Output and Folder Structure

Input Data

- *3D model* of the hand in .OBJ format.
- *Associated texture* in .PNG, .JPG, .JPEG, .TIF, .TIFF, .BMP or .WEBP format.
- *Output folder* where the process will be saved for later reuse, together with the results.
- *CSV file of anthropometric measurement operations*, a table that defines which measurements will be calculated. LandmarksOperations_data.csv is loaded by default and can be replaced with a custom CSV file in Measures selector.

Input best practices. Keeping the OBJ and texture in the same folder helps prevent broken paths. The application tries to detect a texture with the same name as the OBJ automatically.

Main results (*results Folder)*

- hand_rigged_labeled.blend → Final Blender scene with the rig and labels applied.
- hand_measures_detailed.xlsx → Detailed table of the measurements obtained.
- points_xyz → Table that identifies each labeled point in 3D space.

Common Status Values in *hand_measures_detailed.xlsx*

This spreadsheet details the measurements obtained during the program workflow. Each obtained measurement is defined by: TYPE, FINGER, DIMENSION, MARKER1, MARKER2, DIST_EUCLIDEA and STATUS. More information is available in the Measures CSV section. The last two fields identify the numerical result of the operation and its status, respectively.

STATUS can have the following values depending onthe result obtained:

Status	Meaning
OK	The measurement was calculated correctly.
MISSING_MARKER	One of the labels required by the measurement is missing.
INVALID_RECIPE	The measurement definition contains invalid markers or an incompatible recipe.

Preparing the Anatomical Markers

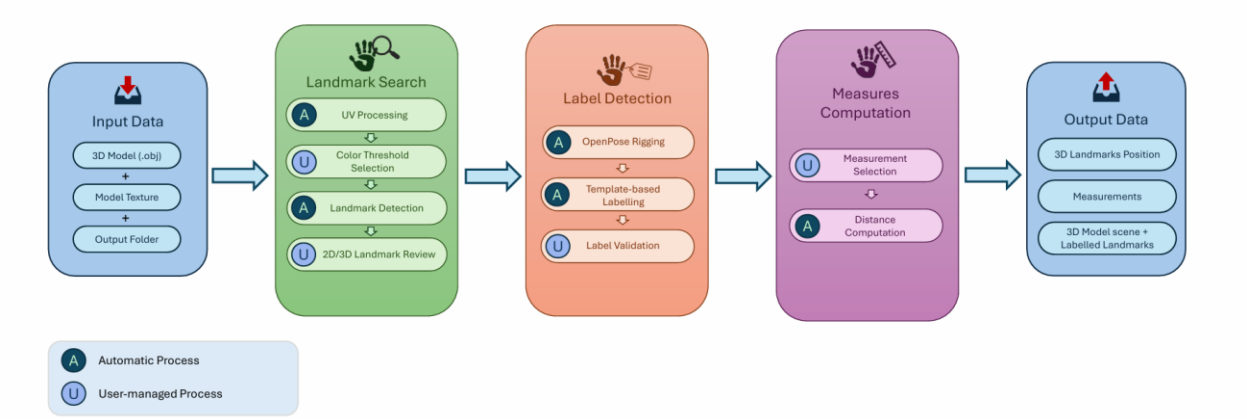
To support automatic color-based detection, it is recommended to use a high-coverage matte blue marker. In the system tests, Catrice Calligraph Artist Matte eyeliner, shade 020 - Ocean Flirt, was applied as small dots on the anatomical references of interest.

Avoid excessively large strokes, shiny areas, irregular stains or partially hidden markers.

4. Quick Guide and Workflow

1. Run ***HAMLET.exe*** from the extracted release folder.
2. In ***Welcome***, start a new measurement with ***New measurement*** or open the output folder (***Open Project*****)** of a previous project.
3. In ***Input Data***, select the OBJ model, the associated texture and the output folder.
4. In ***Color Threshold Selection***, use *Blue matte color preset* if the markers follow the matte blue protocol, or *Manual color selection* if the texture requires threshold.
5. In ***Landmark Detection + 2D/3D Validation***, detect the markers, review their position in 2D and 3D, and correct them before continuing.
6. In ***Rigging and Labeling***, generate the rig, run *OpenPose*, apply automatic labeling and correct the required labels in the interactive editor.
7. In ***Measures selector***, choose the measurements to calculate and add custom measurements if needed.
8. In ***Calculation of measures,*** review the final preview, open the CSV file if desired and press ***Finish*** to access the results folder.

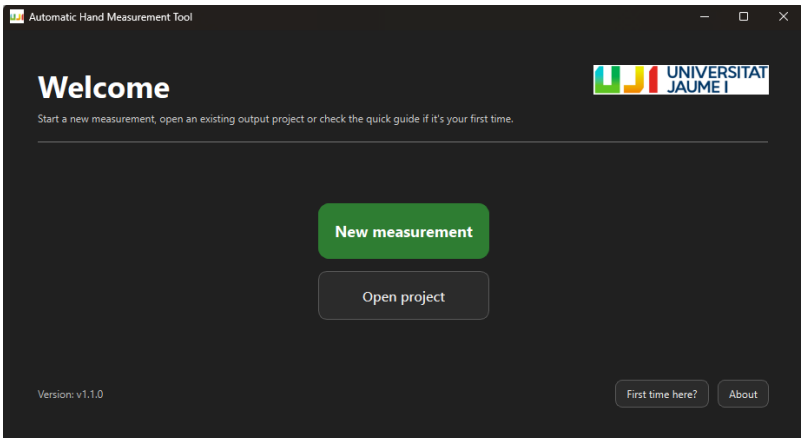
2D/3D review and **manual correction of labels** are the main controls for obtaining reliable measurements.



5. Screen-by-Screen Guide

Welcome

Initial application screen. It allows the user to start a new measurement, open an existing project, consult quick help or view application information.



Control	Function
New measurement	Starts the normal workflow from Input Data.
Open project	Allows the user to select the output folder of a previous project and reopen available parts.
First time here?	Displays a brief guide inside the application.
About	Displays the version, developer and tool description.

Input Data

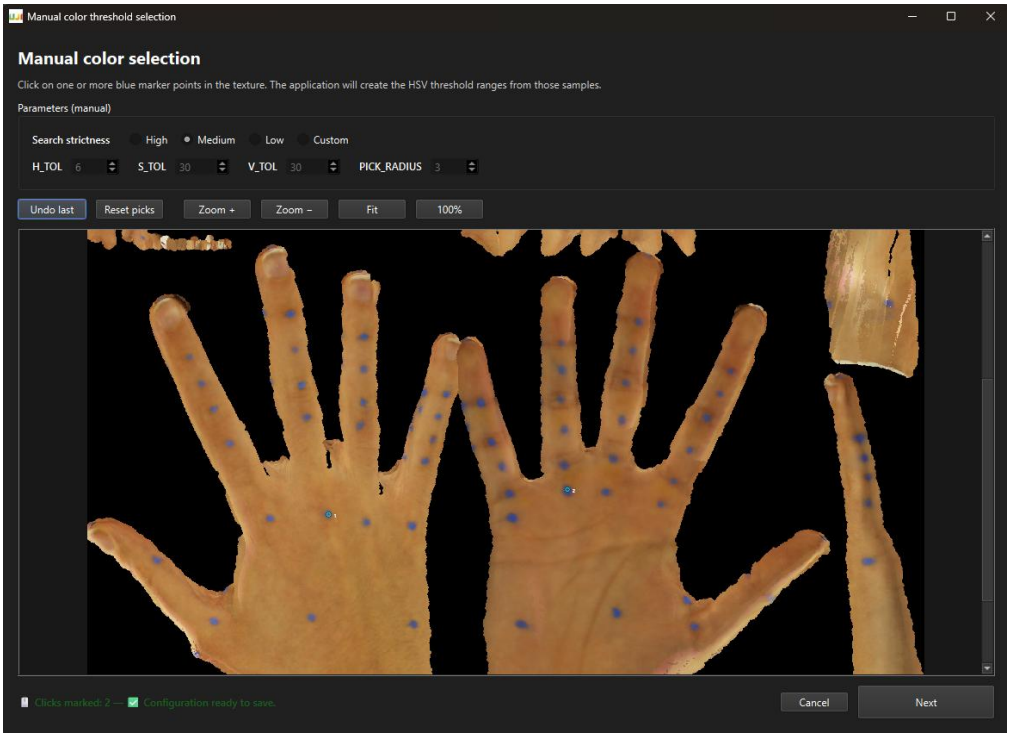
This screen defines the input files and the folder where the results will be generated. The ***Next*** button is enabled only when the OBJ, texture and output folder are valid.

Field	Description
3D Model (.obj)	Geometric model of the hand. It must contain geometry and UV coordinates.
Texture	Image associated with the model. If it has the same name as the OBJ, it can be detected automatically.
Output folder	Folder where CSV files, auxiliary OBJ files, BLEND files and final results are written.

Color Threshold Selection

Defines the parameters used to search for markers on the texture. The recommended mode is Blue matte color preset when the markers follow the protocol color. Manual color selection opens a texture view where the user selects color samples and generates custom HSV ranges.

- High: more restrictive threshold, useful when there are false positives.
- Medium: balanced option, recommended for most manual cases.
- Low: broad tolerance, useful when there are lighting or color variations.
- Custom: allows editing H_TOL, S_TOL, V_TOL and PICK_RADIUS.



Manual selection. Click several times on clean, homogeneous marker areas. Avoid edges, highlights and areas mixed with skin color or shadows.

Landmark Detection + 2D/3D Validation

The application detects markers on the texture, projects them onto the 3D mesh and opens a combined validation step. The screen summarizes two phases: *Automatic Landmark Detection* and *2D/3D Validation.*

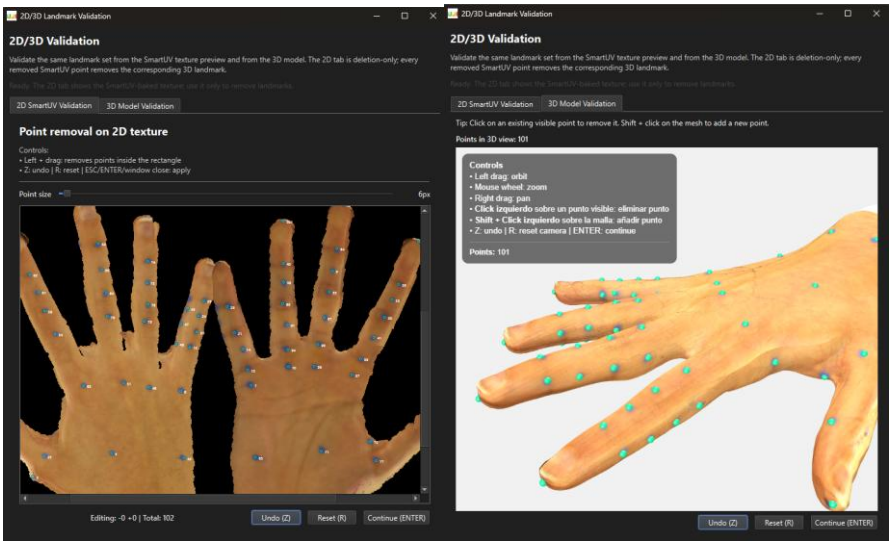
Element	Use
Automatic Landmark Detection	Displays the initial 2D points, the 3D mapping status and the mapped total.
2D/3D Validation	Allows the user to review and edit the same set of landmarks from the texture and from the mesh.
Retry	Repeats automatic detection or only 2D/3D validation, depending on the available information.
Next	Generates hand_with_points.obj and moves on to the rigging and labeling workflow.

2D Editing

- Useful for removing visible false positives from the texture.
- Allows points to be deleted in the 2D image.
- After applying 2D changes, the application synchronizes the 3D CSV when possible.

3D Editing

- Useful for confirming that the points are actually on the surface of the hand.
- ***Shift + click*** on the mesh adds a new point on the surface.
- Clicking an existing point allows it to be deleted if it is visible and selectable.
- ***Undo*** and ***Reset Camera*** help correct editing errors and view orientation.



Rigging and Labeling

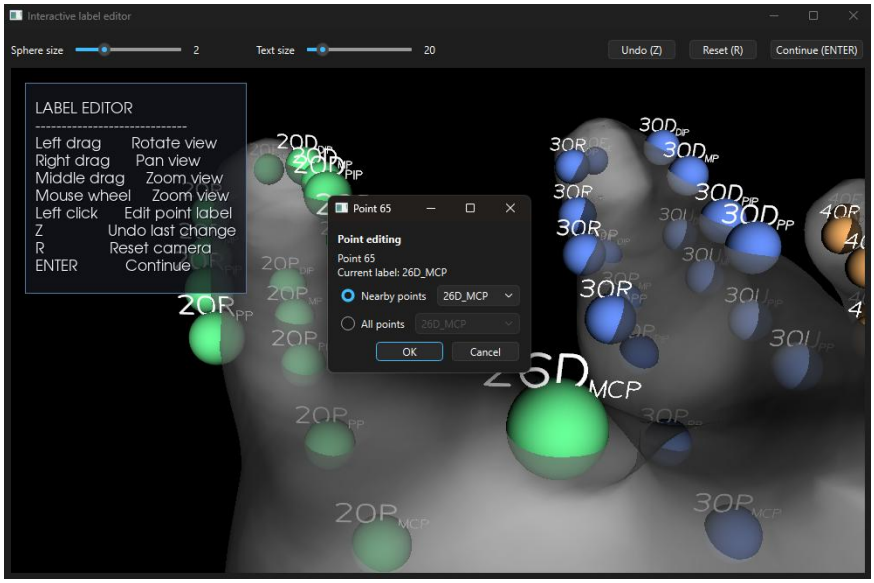
This screen groups *Automatic Rigging*, *Automatic Labeling* and *Interactive Label Editor* into a single guided workflow. The screen displays status cards and allows specific phases to be retried when the required intermediate files exist.

Phase	What it does
Automatic Rigging	Renders a top view with Blender, runs OpenPose on the image and builds the armature in Blender.
Automatic Labeling	Identifies which anatomical point each detected sphere represents by comparing its 3D position with an armature-based anatomical reference, and assigns a unique label to each point.
Interactive Label Editor	Opens a 3D editor for manually reviewing and correcting labels.

Model orientation. If OpenPose or automatic labeling fails generally, review the orientation of the imported model. The workflow assumes an orientation compatible with the render used by rigging.

Interactive Label Editor

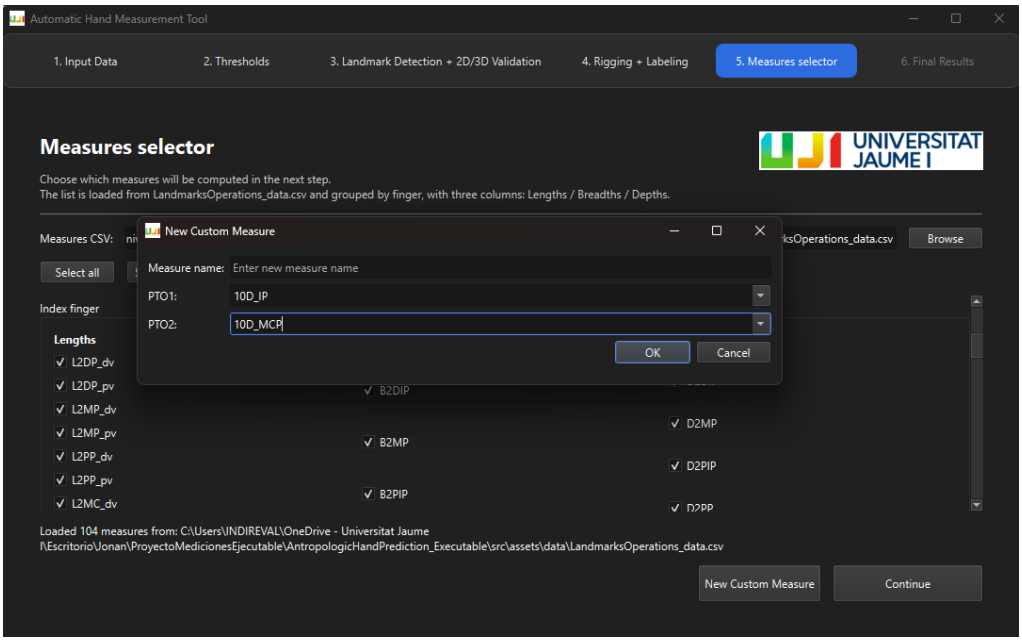
- Left-clicking a point opens the *Point editing*.
- *Nearby points* displays nearby or more likely candidates.
- *All points* allows the user to search for any available label.
- If a label already assigned to another point is selected, the application swaps the labels.
- **ENTER** or **Continue** closes the editor and saves the correction.



Measures selector

Allows the user to choose which measurements to calculate. By default, it loads LandmarksOperations_data.csv from assets/data. Measurements are grouped by finger and type: *Lengths*, *Breadths* and *Depths*. It also allows a custom operations CSV file to be loaded.

Control	Function
Browse	Select a custom operations CSV file.
Select all	Selects all available measurements.
Select none	Clears all selected measurements.
New Custom Measure	Creates an extra measurement between two existing labels.
Continue	Saves LandmarksOperations_selected.csv and proceeds to calculation.



Final Results

Calculates Euclidean distances from the operations selected in the previous step. The screen shows a preview with the measurements and a summary of OK, MISSING and INVALID statuses.

- **Open XLSX** opens the generated results table.
- **Finish** confirms the final results, opens the results folder and closes the application.

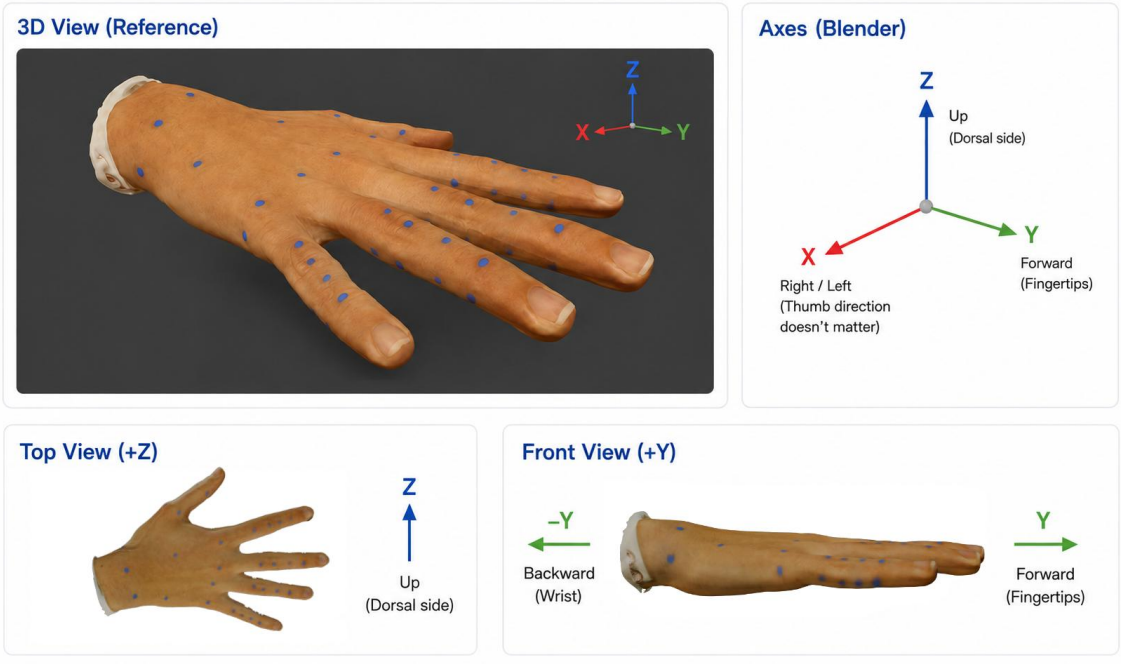
When to Repeat a Step

Situation	Recommended action
Many false positives in the texture	Return to the threshold step and use a stricter preset or more precise manual selection.
Correct points in 2D but poorly located in 3D	Review the 3D tab and add/delete points directly on the mesh.
OpenPose shows an inconsistent structure	Review the model orientation and repeat Automatic Rigging.
Many incorrect labels	Use *Interactive Label Editor* and, if the problem is general, review orientation and rigging.
Measurements with MISSING_MARKER	Check that the required labels exist in detections_labeled.csv.

Required Orientation of the 3D Object for Processing

The 3D model must be exported with an orientation compatible with the top-view render used by the rigging and OpenPose workflow. In the expected reference orientation, the wrist points toward -Y, the fingers extend toward +Y and the back of the hand faces +Z.

If the model is exported rotated or inverted, correct its orientation in Blender and export it again before starting the measurement.



6. Reopening an Existing Project

The ***Open project*** screen allows the user to select an *output folder* created by a previous run. The application inspects the available files and enables only the actions it can safely recover.

Action	Main requirements
Modify markers / detections	Texture, OBJ with UV, detections_all.csv and detections_mapped_3d.csv.
Modify labels manually	OBJ, detections_labeled.csv with labels and XYZ coordinates, and a rigged/labeled .blend file.
Modify measures to obtain	detections_labeled.csv with labels and XYZ coordinates.

Compatibility. If an action appears disabled, hovering over the button may show which file is missing. Recent projects include project_context.json to make reopening easier even if folders have been moved.

7. Measures CSV Format

Operations CSV files define which distances are calculated. They must be semicolon-separated and contain the expected columns. When a row does not repeat the measurement type or finger, the values from the previous row are used. Each measurement, however, must always indicate its dimension and the markers that define it.

Column	Description
TYPE	Measurement category: LENGTHS, BREADTHS, DEPTHS or EXTRAS.
FINGER	Finger or anatomical group associated with the measurement.
DIMENSION	Name of the measurement that will appear in the results.
MARKER 1	Label of the first point used to calculate the distance.
MARKER 2	Label of the second point used to calculate the distance.

Exact match. The values of MARKER *1* y MARKER *2* must match exactly the labels present in detections_labeled.csv.

8. Glossary

Concept	Meaning
Basic concepts	
3D model	Digital representation of the shape of the hand. It can be viewed from different sides while observing the color associated with each point of the corresponding 3D mesh.
OBJ	File type that stores the shape of the 3D model and the information required to correctly place its color image.
2D texture	Flat image that provides the color and visible details of the 3D model. The anatomical markers must be clearly visible on it.
Anatomical marker / Landmark	Small colored dot drawn on the hand to indicate a point of interest used in a measurement.
Label	Name assigned to a landmark to indicate the specific location on the hand it represents.
Rigging	Process through which the application places a virtual guide inside the hand to recognize the arrangement of the wrist and fingers.
Measures CSV	Table-like file containing the definition of each measurement (the points used, the measurement type and the anatomical area it belongs to).
Euclidean distance	Straight-line separation between two points. It is the numerical value used by the application to obtain the measurements.
Additional concepts	
HSV threshold	Range of color hue (H), saturation (S) and brightness/value (V) that the application accepts to decide which areas of the 2D texture correspond to anatomical markers.
OpenPose	Component that analyzes an image of the hand and estimates where the wrist and finger joints are located.
Blender	3D creation program that the application uses automatically to generate views of the hand, place the internal guide and save the final result.
Render	Image created from the 3D model from a specific point of view. OpenPose uses this image to analyze the hand.

Concept	Meaning
Armature	Virtual skeleton made of connected parts and placed inside the 3D model. It serves as a reference for assigning labels.
.exe file	Windows file that launches the application when double-clicked.
Files and results	
detections_all.csv	List of the points initially found on the 2D texture before checking their position on the 3D model.
detections_mapped_3d.csv	List of the points that could be mapped from the 2D texture to 3D space, together with their exact position.
detections_labeled.csv	Final list of reviewed landmarks, with their label and position. The application uses this file to calculate measurements.
hand_with_points.obj	Copy of the 3D hand model showing small spheres at the detected positions. It is used in later steps of the process.
hand_rigged_labeled.blend	Final Blender file containing the hand, the internal guide and the points with their labels.
points_xyz.xlsx	Spreadsheet containing each labeled point and its X, Y and Z values, which indicate its position in 3D space.
hand_measures_detailed.xlsx	Detailed table of the calculated measurements. It indicates the points used, the value obtained and the status of each result.
Statuses and errors	
OK	The measurement was calculated correctly.
MISSING_MARKER	One of the labeled points required to calculate the measurement was not found.
INVALID / INVALID_RECIPE	The measurement cannot be calculated because its definition is incomplete or contains a point combination that is not allowed.

User Responsibilities

- Check that the OBJ and texture belong to the same model and are correctly linked.
- Visually verify that the detected points correspond to real markers.
- Correct points or labels when automatic detection is not effective.
- Review the results table (Output file) before using the measurements obtained in external analyses.

The application automates much of the process, but the final quality of the anatomical dimensions obtained depends on reviewing the landmarks and labels before calculating measurements. Visual validation is essential for obtaining reliable results.

9. FAQ and Troubleshooting

Problem	Recommended action
The application does not start on another computer	Check that the complete release folder was copied, not only HAMLET.exe. It is also advisable to extract the ZIP file to a local path with write permissions.
Blender cannot be found or access is denied	The release must contain blender/blender.exe. If a manual path is configured, it must point to the blender.exe executable or to the folder that contains it.
OpenPose does not detect the hand correctly	Review the model orientation, the render quality and the presence of openpose/models/hand/pose_deploy.prototxt and pose_iter_102000.caffemodel.
Too many false positives are detected	Use a more restrictive threshold, reduce HSV tolerances or select cleaner and more homogeneous manual samples.
Too few points are detected	Use broader tolerances, select more samples in different marker areas and avoid highlights or extreme shadows.
A point appears in 2D but does not transfer correctly to 3D	Not all texture points can be projected onto the mesh if the UV mapping or geometry does not allow it. Always validate the 3D tab before continuing.
Automatic labeling is mostly incorrect	The most common cause is an incompatible model orientation or an incorrect rig. Review the OpenPose result and correct labels manually.
A custom measurement cannot be added	Check that the name is not empty, that no other measurement has the same name and that PTO1 and PTO2 are different.
The custom measures CSV does not load	Check the semicolon separator, compatible encoding and the TYPE, FINGER, DIMENSION, MARKER 1 and MARKER 2 columns.
The final result shows MISSING_MARKER	One of the labels required by the measurement does not exist in detections_labeled.csv or was left empty after manual correction.

10. Contact

For questions about using the application, incidents during the measurement workflow or clarification about the generated results, contact the document owner or the team responsible for maintaining the tool.

Reference	Details
Owner	Jon Andoni Fernández Moyano (moyanoj@uji.es), Universitat Jaume I (UJI), Castelló de la Plana, Spain
Recommended query details	Include the application version, the project folder used and a brief description of the problem or question.